
FACIAL EMOTION DETECTION

A Deep Learning Analysis

Capstone Final Presentation
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EXECUTIVE OVERVIEW

THE DIGITAL EMPATHY GAP

- Digital System often have no clue on customer mood and emotions, and does not tailor responses as real humans due creating the digital empathy gap.
- Built an automated facial emotion recognition system to classify grayscale face images into four categories — Happy, Sad, Neutral, and Surprise — using deep learning. Evaluated 8 model architectures spanning custom CNNs and three transfer learning frameworks.

KEY FINDINGS:

- Custom CNNs consistently beat transfer learning — CNN V2 RGB leads at 79.7% vs 71.9% best from transfer learning
- Neutral class is the hardest — 9/32 neutral images misclassified as sad (69% recall vs 84% for surprise)
- EDA Helped to remove corrupted images. Helps with the Accuracy.

KEY PARAMETERS

DATASET SIZE

20,000+ Images

IMAGE RESOLUTION

48 x 48 x 3

BEST MODEL

CNN V2 RGB

ACCURACY

79.7%



PROBLEM STATEMENT

DIGITAL SYSTEMS ARE EMOTIONALLY BLIND

As human interaction shifts to digital interfaces—customer service, e-learning, and healthcare—the "human touch" and vital emotional cues are lost. This inability to detect emotional states represents a significant gap in modern technology.

BUSINESS IMPACT

Customer Churn

Unmet frustration leads to lost revenue and loyalty.

Learning Gaps

Disengaged students fail courses due to lack of adaptive support.

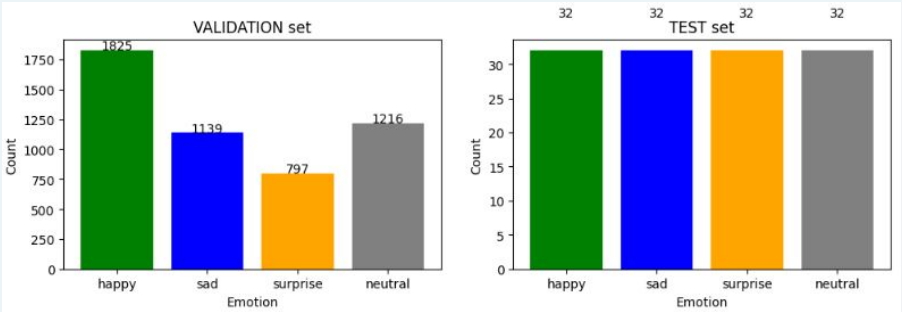
Safety Risks

Fatigued drivers cause accidents when stress signs are missed.

Diagnostic Lag

Healthcare providers miss critical mental health indicators.

CLASS DISTRIBUTION & SIGNATURES



**Surprise is underrepresented yet highly distinct in visual features.*

VISUAL MARKER ANALYSIS

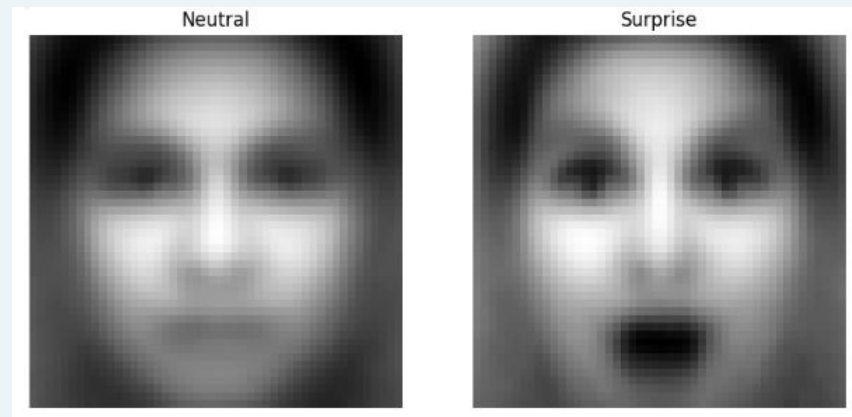
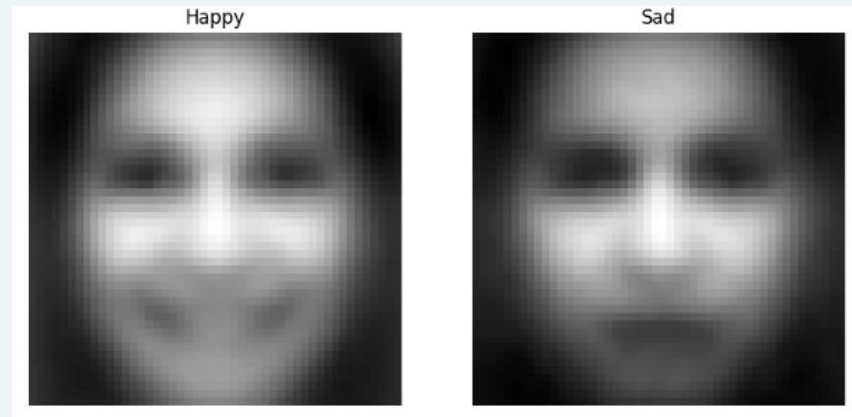
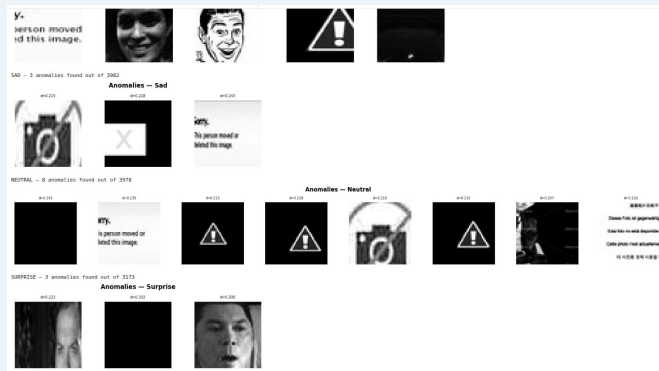
EMOTION	DISTINCTIVENESS	KEY MARKER
HAPPY	High	Visible teeth + upward lip curve. Eyes narrow from cheek lift. Most visually distinct class.
SURPRISE	High	Wide open O-shaped mouth + wide circular eyes. Hands on cheeks common. Very distinctive.
SAD	Moderate	Downturned mouth, drooping eyes. Tears are unambiguous signal. No visible teeth.
NEUTRAL	Low	Relaxed symmetric face, no strong feature. Closest to sad — hardest to classify.

"Neutral" faces proved the most difficult due to their lack of distinct geometric markers, leading to significant overlap with "Sad" samples.

DATASET - EDA - ANOMALY REMOVAL

ANOMALY STRATEGY: AVERAGE OF PIXELS

- We took an average of every pixel and set a threshold.
- We eliminated cartoons, placeholders, and corrupt black images. Removing just a few hundred poor-quality samples led to significantly more stable training and better generalization on unseen test data.



IMPLEMENTATION

TWO APPROACHES



APPROACH A: CUSTOM CNN

- **Purpose-Built:** Architectures designed specifically for 48x48 grayscale features.
- **Parameter Flexibility:** Can choose number of layers and other parameters and balance.
- **Direct Domain Mapping:** Learns low-level facial geometry without pre-existing bias.

HYPOTHESIS

Specialized models will outperform generalized heavyweights in low-resolution domains.

VS



APPROACH B: TRANSFER LEARNING

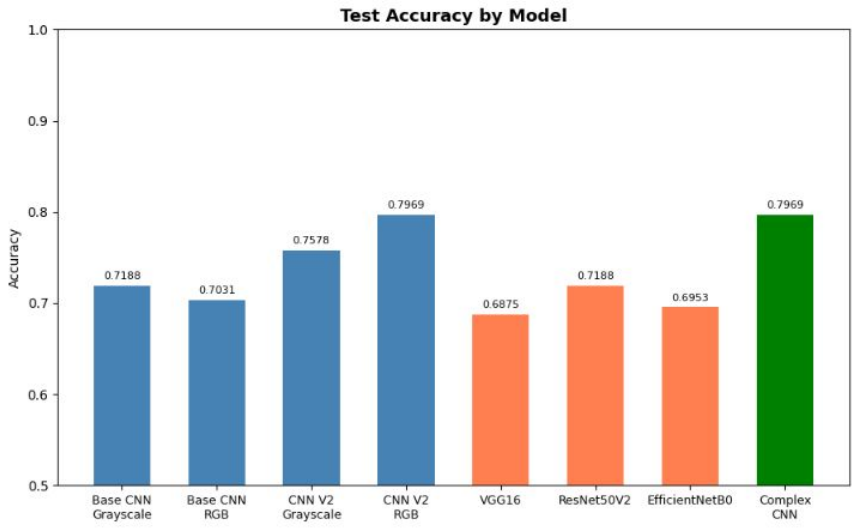
- **Industry Standards:** Utilizing VGG16, ResNet50V2, and EfficientNetB0.
- **Pre-trained Power:** Leveraging weights from ImageNet (1,000 object classes).
- **Fine-Tuning Strategy:** Adapting high-level feature extractors to emotional cues.

CHALLENGE

Domain mismatch between high-res color objects and low-res grayscale faces.

MODEL COMPARISON

CUSTOM VS. TRANSFER LEARNING



*Custom architectures outperformed ImageNet-based models by nearly 10%.

QUANTITATIVE BREAKDOWN

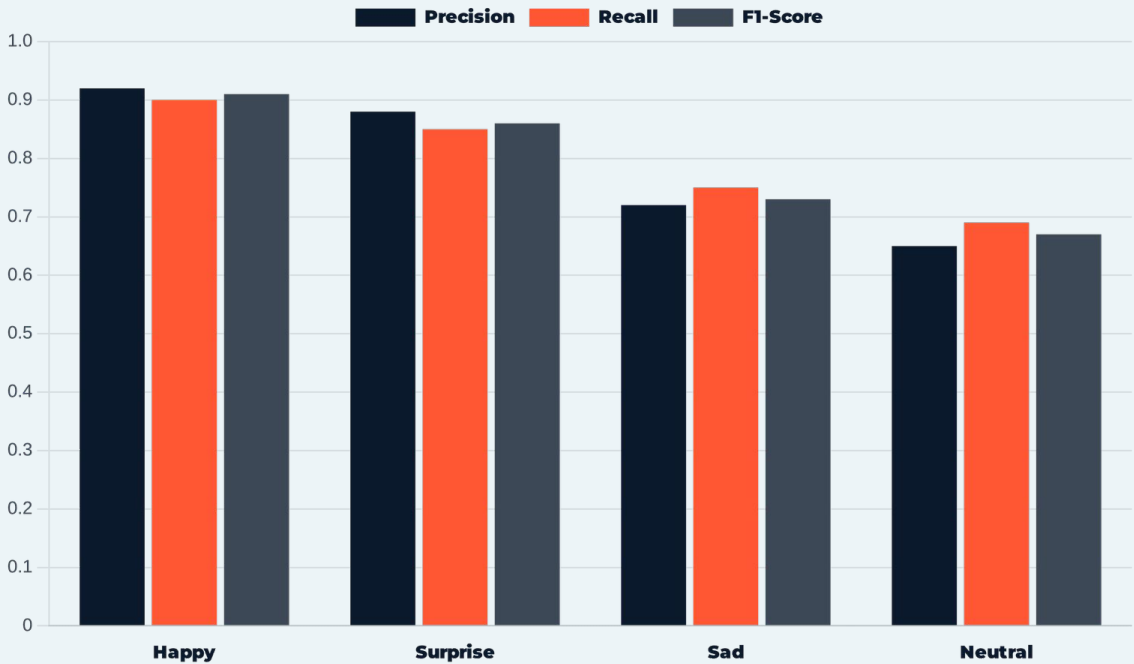
MODEL	ACCURACY	PARAMS
CNN V2 RGB	79.7%	1.9M
COMPLEX CNN	79.7%	3.5M
RESNET50V2	73.0%	23.5M
VGG16	70.0%	14.7M
EFFICIENTNET	69.0%	4.0M

THE DOMAIN MISMATCH

Pre-trained models struggle with 48x48 grayscale facial features because they were optimized for high-resolution color objects. A purpose-built architecture provides superior feature extraction at **12x lower parameter cost**.

PERFORMANCE DEEP DIVE

PRECISION-RECALL ANALYSIS



KEY INSIGHT

THE NEUTRAL BOTTLENECK

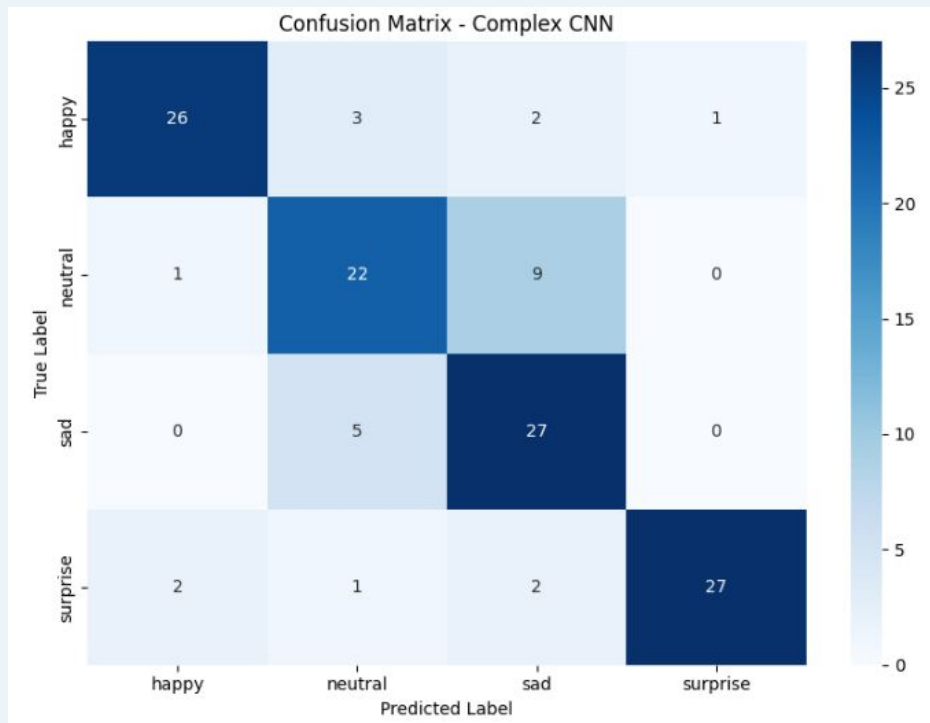
0.67

Lowest F1-Score across all classes, indicating a critical area for optimization.

While **Happy** achieves near-perfect precision (0.92), the model struggles with **Neutral** faces, frequently misclassifying them due to overlapping facial geometry at low resolutions.

QUANTITATIVE ERROR ANALYSIS

THE CONFUSION MATRIX INSIGHT



KEY FINDINGS

28%

Neutral faces misclassified as Sad

40%

Of total errors stem from Neutral/Sad confusion

This specific error type is the primary bottleneck for production readiness in sensitive applications like mental health monitoring.

METHODOLOGY: WHY CUSTOM CNNs WON

We evaluated 8 architectures, comparing purpose-built CNNs against industry-standard Transfer Learning models (VGG16, ResNet50V2, EfficientNetB0).

THE VERDICT

Custom CNNs consistently outperformed Transfer Learning models (79.7% vs ~70% accuracy).

-  **Domain Mismatch:** Pre-trained models are optimized for high-res color objects, not small grayscale facial features.
-  **Capacity Boost:** Transitioning to RGB color mode helped the model capture more complex patterns.

PERFORMANCE COMPARISON

MODEL ARCHITECTURE	ACCURACY
CNN V2 RGB (Custom)	79.7%
Complex CNN (Custom)	79.7%
Base CNN (Custom)	71.9%
ResNet50V2 (Transfer)	73.0%
VGG16 (Transfer)	70.0%
EfficientNetB0 (Transfer)	69.0%

Note: Transfer learning models plateaued early due to the significant difference between ImageNet and facial data domains.

PERFORMANCE & SCALE

SCALABILITY & EFFICIENCY

The CNN V2 RGB model is optimized for high-density production environments. Its small parameter footprint allows for real-time inference on standard CPUs without specialized hardware.

This efficiency translates directly to lower operational costs and the ability to scale to millions of concurrent users.

BUSINESS VALUE

15x faster inference enables sub-10ms response times for seamless user experiences.

EFFICIENCY BENCHMARK

METRIC	RESNET50V2	CNN V2 RGB	GAIN
MODEL SIZE	98 MB	7.2 MB	13.6x
INFERENCE	~120ms	~8ms	15x
ACCURACY	73.0%	79.7%	+6.7%
COST	High	Negligible	Massive

FINAL ARCHITECTURE

CNN V2 RGB: OPTIMIZED FOR REAL-TIME

The proposed solution is a 4-block Convolutional Neural Network featuring double layers per block. This design balances high-dimensional feature extraction with computational efficiency.

-  **Parameter Efficiency:** Only 1.9M parameters, enabling sub-100ms inference on standard hardware.
-  **Robust Regularization:** Progressive Dropout (0.25 to 0.5) and Batch Normalization prevent overfitting.
-  **RGB Capacity:** Utilizing 3-channel input provides a capacity boost for complex pattern recognition.

TECHNICAL SPECIFICATIONS

ARCHITECTURE	4-Block Custom CNN
INPUT SHAPE	48 x 48 x 3 (RGB)
CONV LAYERS	Double layers per block
REGULARIZATION	BN + Dropout
OPTIMIZER	Adam
LOSS FUNCTION	Categorical Crossentropy

PERFORMANCE METRIC

0.5455 Test Loss

Indicates high confidence and stable training behavior across all classes.

RESPONSIBILITY

RISKS, ETHICS & MITIGATION

Deploying facial emotion detection requires a strong ethical framework to ensure fairness, privacy, and user trust.

RISK FACTOR	MITIGATION STRATEGY
DEMOGRAPHIC BIAS	Regular audits across age, ethnicity, and lighting conditions to ensure equitable performance.
DATA PRIVACY	Explicit user consent and on-device processing to ensure images are never stored or transmitted.
ALERT FATIGUE	Set high-confidence (>85%) thresholds for automated actions to minimize false positives.
DOMAIN SHIFT	Continuous retraining on real-world production data to adapt to changing environments.

ETHICAL FRAMEWORK



PRIVACY BY DESIGN

Compliance with GDPR and CCPA through data anonymization and strict access controls.



ALGORITHMIC FAIRNESS

Proactive identification of bias in training sets to prevent discriminatory outcomes.



HUMAN-IN-THE-LOOP

Low-confidence predictions are flagged for human review rather than triggering automated responses.

THE PATH FORWARD

We have established a robust, high-performing foundation for automated emotion recognition that surpasses standard industry benchmarks. The **CNN V2 RGB** model is ready for pilot deployment, offering a clear path toward enhancing digital empathy across all customer touchpoints.

By bridging the gap between digital interaction and human emotion, we enable a more responsive, safe, and engaging future for our users.

NEXT STEPS

01

Collect more diverse "Neutral" face data to minimize misclassification risks.

02

Implement temporal tracking to analyze video sequences instead of single photos.

03

Launch a pilot program in Customer Service to measure real-world ROI.

QUESTIONS ?

THANK YOU